

Curriculum Vitae

Xin (Selena) Feng

Contact Details

Department of Geography and Environmental Sustainability
University of Oklahoma, Norman, OK 73019, USA
Email: selena.feng@ou.edu

Teaching and Research Interests

Geographic information science; spatial optimization; GeoAI; urban, regional, and natural resource planning and development; severe weather observation; emergency response; infrastructure and transportation systems.

Education

Ph.D., Geography (Spatial Analysis and Modeling) University of California, Santa Barbara, USA	2019
M.A., Geographical Sciences and Urban Planning Arizona State University, USA	2015
M.S., Remote Sensing & GIS Peking University, China	2013
B.S., Cartography and Geographical Information System Wuhan University, China	2010

Academic Appointments

2021-present	Tenure-Track Assistant Professor, Department of Geography and Environmental Sustainability, University of Oklahoma
2020-2021	Postdoctoral Scholar, Center for Geospatial Sciences, University of California, Riverside
2016-2019	Research Assistant and Lab Lecturer, Department of Geography, University of California, Santa Barbara
2013-2015	Research Assistant and Teaching Assistant, School of Geographical Sciences and Urban Planning, Arizona State University

Publications

Underlined authors are my student advisees.

Refereed Journal Articles Under Review

- B. Parajuli, **X. Feng**. Continuous Multi-Facility Coverage Location Problem with Inter-Facility Geometric Separation and Obstacle-Constrained Service Coverage. *Journal of*

Geographical Systems. (Submitted November 2025; major revision requested in January and June 2026.)

Published Refereed Journal Articles

- 2026 **X. Feng**, Y. Cao. A Benchmark of Large Language Models for Spatially Informed Proposal Generation in Local Search for the p-Median Problem. *Computers, Environment, and Urban Systems*. 129, 102467. <https://doi.org/10.1016/j.compenvurbsys.2026.102467>.
- 2026 K. Wu, X. Ye, S. Jamonnak, **X. Feng**. Human Mobility Reimagined: Digital Twin Intelligence for Adaptive Campus Course Timetabling. *Transactions in GIS*. Accepted.
- 2026 C. Wang, **X. Feng**, A. A. Alford. Optimizing Dual-Doppler Mobile Radar Siting with an Obstacle-Aware Backup Coverage Model for Strategic Storm Data Collection. *IEEE Transactions on Radar Systems*, 4, 841-854. <https://doi.org/10.1109/TRS.2026.3682657>.
- 2025 R. Wei, **X. Feng**, S. Rey. Delineating School Attendance Boundaries to Improve Diversity and Community Integrity. *Environment and Planning B: Urban Analytics and City Science*, 23998083251389075. <https://doi.org/10.1177/23998083251389075>.
- 2025 **X. Feng**, Y. Cao. Enhancing Accessibility of Regionalization Techniques through Large Language Models: A Case Study in Conversational Agent Guidance. *International Journal of Geographical Information Science*, 1–24. <https://doi.org/10.1080/13658816.2024.2415439>.
- 2025 B. Parajuli, **X. Feng**. Deployment Optimization of a Mobile Weather Radar: A Cover-Based Single Facility Location Problem in a Constrained Continuous Space. *Remote Sensing*, 17(5), 870. <https://doi.org/10.3390/rs17050870>.
- 2024 **X. Feng**, D. Schwartzman, B. Parajuli, X. Wang. Enhancing Meteorological Mobile Radar Observations through Deployment Location Optimization. *IEEE Transactions on Geoscience and Remote Sensing*, 62, 1-11. <https://doi.org/10.1109/TGRS.2024.3447550>.
- 2024 M. S. Parvez, **X. Feng**. A Generic Method for Social–Environmental System Boundary Delineation: An Amalgamation of Spatial Data Integration, Optimization, and User Control for Resource Management. *ISPRS International Journal of Geo-Information*, 13(12), 447. <https://doi.org/10.3390/ijgi13120447>.
- 2024 **X. Feng**, J. Koch. Combining Vector and Raster Data in Regionalization: A Unified Framework for Delineating Spatial Unit Boundaries for Socio-Environmental System Analysis. *International Journal of Applied Earth Observation and Geoinformation*, 128, 103745. <https://doi.org/10.1016/j.jag.2024.103745>.
- 2023 **X. Feng**. Multi-Plant Location Involving Resource Allocation. *Geographical Analysis*, 56(1), 97-117. <https://doi.org/10.1111/gean.12366>.
- 2023 Y. Zhu, S. W. Myint, **X. Feng****, & Y. Li. An Innovative Scheme to Address the Trade-Off between Water Conservation and Heat Alleviation with Environmental Justice for Urban Sustainability: The Case of Phoenix, Arizona. *AGU Advances*, 4(1), e2022AV000816. (**corresponding author. This paper was selected for featuring as an Editor’s Highlight, which was published on Eos.org)
- 2022 R. Wei, **X. Feng**, S. Rey, & E. Knaap. Reducing Racial Segregation of Public School Districts. *Socio-Economic Planning Sciences*, 84, 101415. <https://doi.org/10.1016/j.seps.2022.101415>.

- 2022 **X. Feng**, G. Barcelos, J. D. Gaboardi, E. Knaap, R. Wei, L.J. Wolf, ... & S. J. Rey. Spopt: A Python Package for Solving Spatial Optimization Problems in PySAL. *Journal of Open Source Software*, 7(74), 3330. <https://doi.org/10.21105/joss.03330>.
- 2022 **X. Feng**, S. Rey, & R. Wei. The Max-p-Compact-Regions Problem. *Transactions in GIS*, 26(2), 717-734. <https://doi.org/10.1111/tgis.12874>.
- 2021 **X. Feng**, A. Murray, R. Church. Drone Service Response: Spatiotemporal Heterogeneity Implications. *Journal of Transport Geography*, 93, 103074. <https://doi.org/10.1016/j.jtrangeo.2021.103074>.
- 2021 **X. Feng**, S. Wang, A. Murray, Y. Cao, S. Gao. MOTO: A Multi-Objective Trajectory Optimization Method for Finding Sequential Activity Locations over Space and Time. *Environment and Planning B-Urban Analytics and City Science*, 48(4), 945-963. <https://doi.org/10.1177/2399808320913300>
- 2020 **X. Feng**, A. Murray. Spatiotemporally Heterogeneous Allocation to Support Service Area Response. *Computers, Environment, and Urban Systems*, 82, 101503. <https://doi.org/10.1016/j.compenvurbsys.2020.101503>
- 2020 A. Murray, R. Church, **X. Feng**. Single Facility Siting Involving Allocation Decisions. *European Journal of Operational Research*, 284 (3), 834-846.
- 2018 **X. Feng**, A. Murray. Allocation Using a Heterogeneous Space Voronoi Diagram. *Journal of Geographical Systems*, 20(3), 207-226.
- 2018 S. Wang, S. Gao, **X. Feng**, A. Murray, Y. Zeng. A Context-Based Geoprocessing Framework to Find Optimal Spatiotemporal Meetup Location on Road Networks for Multiple Moving Objects. *International Journal of Geographical Information Science*, 32(7), 1368-1390. <https://doi.org/10.1080/13658816.2018.1431838>.
- 2018 **X. Feng**, A. Murray. Spatial Analytics for Enhancing Street Light Coverage of Public Spaces. *LEUKOS*, 14(1): 13-23. <https://doi.org/10.1080/15502724.2017.1321486>.
- 2016 A. Murray, **X. Feng**. Public Street Lighting Service Standard Assessment and Achievement. *Socio-Economic Planning Sciences* 53: 14-22. <https://doi.org/10.1016/j.seps.2015.12.001>.
- 2016 **X. Feng**, S. Myint. Exploring the Effect of Neighboring Land Cover Pattern on Land Surface Temperature of Central Building Objects. *Building and Environment*, 95: 346-354.
- 2014 J. Song, S. Du, **X. Feng** and L. Guo. The Relationships between Landscape Compositions and Land Surface Temperature: Quantifying Their Resolution Sensitivity with Spatial Regression Models. *Landscape and Urban Planning*, 123: 145-157.
- 2013 **X. Feng**, S. Du, F. Zhang and S. Wang. Urban Land Classification of High-Resolution Images Based on Multi-Scale Fusion. *Journal of Geography and Geo-Information Science* 29(3): 43-47.
- 2011 **X. Feng**, S. Du, H. Shu. The Effect of Spatial Weight Matrices on Spatial Autocorrelation: A Case Study of Hemorrhagic Fever with Renal Syndrome (HFRS) in China. *Geomatics and Information Science of Wuhan University*, 36(12): 1410-1413.

Refereed Conference Proceedings

- 2018 A. Murray, **X. Feng**, Ali Shokoufandeh. Heterogeneous Skeleton for Summarizing Continuously Distributed Demand in a Region. In *Proceedings of 10th International Conference on Geographic Information Science*, vol. 114.
- 2011 **X. Feng**, S. Du, H. Shu. Spatial Regression Analysis in Hemorrhagic Fever with Renal Syndrome (HFRS) in China. *Proceedings of 2011 IEEE International Conference on Spatial Data Mining and Geographical Knowledge Services*, 77-80.

Research Funding

Pending

H. Luan (MPI), **X. Feng (MPI & Institutional PI)**, W. Shi, P. Sullivan. Revolutionizing the Measurement of Spatial Accessibility to Pre-Exposure Prophylaxis Using Large Language Models. NIH NLM R21. \$466,366. 2026-2028. (Under active NLM FY26 funding consideration; Just-in-Time materials requested in June 2026.)

Funded

O. Rowland, **X. Feng (co-PI)**, T. Fagin. Assessment of the Implications of Wildfire–Electric Infrastructure Interactions for Socio-Ecological Resilience in Great Plains Grassland Systems. GRAS SES Seed Grant. \$5,000. 2026-2027.

X. Feng (PI). From Targets to Tracks: A Route-Optimization Prototype for NURTURE. NASA Oklahoma EPSCoR Travel Grant. \$2,924. 2025-2026.

X. Feng (PI), D. Schwartzman. Optimizing the location of Mobile Radar Deployment to Support Meteorological Observation. The University of Oklahoma (Vice President for Research and Partnerships, Faculty Investment Program), \$ 9,981.3. 2023-2024.

X. Feng (PI), J. Koch. Spatially explicit delineation for boundaries of social-environmental systems: a case in the Rio Grande – Rio Bravo basin. The University of Oklahoma Data Institute for Societal Challenges (DISC) Seed Grant, \$10,000. 2023-2024.

Invited Talks (*My Student Advisees*)

X. Feng. Location Modeling for Decision Support: A Case Study in Mobile Radar Deployment. OU Data Institute for Societal Challenges Symposium Presentation. April 2026.

X. Feng. “Location, Location, Location” in Earth Science: Siting Sensors, Routing Platforms, and Large Language Model-Guided Decision Support. NASA Ames Earth Science Division. March 2026.

X. Feng, O. Rowland. Energy Resource Planning under Extreme Weather Conditions. OG&E Transmission and Distribution (T&D) Technology Group. October 2025.

X. Feng, C. Wang, B. Parajuli. Spatial Optimization for Severe Storm Observations: From Mobile Radars to Drones. NOAA National Severe Storms Laboratory Engineering & Observations Team. September 2025.

X. Feng. Enhancing Decision-Making: Integrating Geospatial Analytics and Spatial Optimization into the Design, Deployment, and Evaluation of Natural Hazard Mitigation Strategies. OU Strategic Environmental Research and Development Program (SERDP) and the Environmental Security Technology Certification Program (ESTCP). May 2025.

X. Feng. Spatial Optimization: Bridging Operations Research and Geographic Information Science. 2024 Operation Research Youth Class, Chinese Academy of Sciences. October 2024.

X. Feng. Optimization in Geospatial Data Science. OU School of Industrial and Systems Engineering, Graduate Seminar Series. September 2022.

X. Feng. Spatial Optimization in Geography. OU School of Geosciences, Geosciences Colloquium. October 2022.

Meetings and Conference Presentations (*My Student Advisees*)

- 2026 X. Feng and Y. Cao. “Leveraging LLMs as Intelligent Proposers in Local Search for the p-Median Problem.” American Association of Geographers Annual Meeting, San Francisco, CA, USA, March 17–21, 2026.
- 2026 C. Wang and X. Feng. “Optimal Mobile Radar Deployment Under Stochastic Storm-Track Uncertainty.” American Association of Geographers Annual Meeting, San Francisco, CA, USA, March 17–21, 2026.
- 2026 B. Parajuli and X. Feng. “Siting Observation Facilities in Continuous Space: A Multi-Facility Location Problem with Obstacle-Constrained Overlapping Coverage and Geometric Separation.” American Association of Geographers Annual Meeting, San Francisco, CA, USA, March 17–21, 2026.
- 2025 X. Feng. “A Chain-of-Thought Large Language Model Approach for Accessible Spatial Optimization.” International Association of Chinese Professionals in Geographic Information Sciences, Jiaozuo, China, June 16–19, 2025.
- 2025 X. Feng. “Enhancing Accessibility of Spatial Optimization Techniques Through Large Language Models: A Case Study in Autonomous Agent Guidance for Regionalization.” American Association of Geographers Annual Meeting, Detroit, MI, USA, March 24–28, 2025.
- 2025 C. Wang and X. Feng. “Spatial Analytics for Determining Mobile Radar Deployment Location in Severe Weather Observations.” American Association of Geographers Annual Meeting, Detroit, MI, USA, March 24–28, 2025.
- 2025 M. Parvez, X. Feng. “A Framework for Social-Environmental System Boundary Delineation: Enhancing Quantitative Geography through Spatial Optimization.” American Association of Geographers Annual Meeting, Detroit, MI, USA, March 24–28, 2025.
- 2024 B. Parajuli and X. Feng. “Strategic Deployment of a Single Mobile Weather Radar for the Enhancement of Meteorological Observation: A Coverage-Based Location Problem.” AGU Annual Meeting, Washington, DC, USA, December 9–13, 2024.
- 2024 X. Feng and Y. Cao. “RegionDefiner: Democratizing Regionalization Algorithms Through Large Language Models.” 71st North American Meetings of the Regional Science Association International, New Orleans, LA, USA, November 13–16, 2024.
- 2024 X. Feng and J. Koch. “Delineating Spatial Unit Boundaries for Socio-Environmental Systems Analysis and Modeling: Combining Vector and Raster Data.” American Association of Geographers Annual Meeting, Honolulu, HI, USA, April 16–20, 2024.

- 2024 X. Feng, D. Schwartzman, B. Parajuli, and X. Wang. "Location Optimization of Mobile Radar Deployment." 104th Annual Meeting of the American Meteorological Society, Baltimore, MD, USA, January 28–February 1, 2024.
- 2023 M. Parvez and X. Feng. "Defining the Basic Units of Social-Environmental Systems for Landscape Planning: An Approach Toward Regionalization." Southwest Division of the American Association of Geographers Annual Meeting, Laredo, TX, USA, November 2–3, 2023.
- 2023 B. Parajuli and X. Feng. "A Cover-Based Single Facility Location Problem with Continuous Heterogeneous Demand." 70th North American Meetings of the Regional Science Association International, San Diego, CA, USA, November 15–18, 2023.
- 2023 X. Feng, D. Schwartzman, and B. Parajuli. "A United Framework for Siting a Mobile Radar Vehicle for Meteorological Observation." 70th North American Meetings of the Regional Science Association International, San Diego, CA, USA, November 15–18, 2023.
- 2023 X. Feng. "A Multi-Source, Raw Material Allocation Form of Multi-Plant Weber Problem." American Association of Geographers Annual Meeting, Denver, CO, USA, March 23–27, 2023.
- 2020 X. Feng, S. Rey, and R. Wei. "Max-p-Compact-Regions Problem." 67th North American Meetings of the Regional Science Association International, Virtual Meeting, November 9–13, 2020.
- 2019 X. Feng, A. T. Murray, and R. Church. "Medical Drone Service Response: Spatiotemporal Heterogeneity Implications." American Association of Geographers Annual Meeting, Washington, DC, USA, April 3–7, 2019.
- 2018 X. Feng and A. T. Murray. "Addressing Spatial Allocation: Planning for Drone-Based Emergency Medical Service." Western Regional Science Association 57th Annual Meeting, Pasadena, CA, USA, February 11–14, 2018.
- 2017 X. Feng, S. Wang, S. Gao, and Y. Cao. "Spatial Analytics for Supporting Meetup Location." 64th North American Meetings of the Regional Science Association International, Vancouver, BC, Canada, November 8–11, 2017.
- 2017 X. Feng and A.T. Murray. "Heterogeneous Voronoi diagram." 2017 Annual Meeting of the American Association of Geographers, Boston, MA, USA, April 5-9, 2017.
- 2016 X. Feng and A. T. Murray. "Heterogeneous Space Facility Siting in Order to Maximize Coverage." 63rd North American Meetings of the Regional Science Association International, Minneapolis, MN, USA, November 9–12, 2016.
- 2016 X. Feng and A. T. Murray. "Spatial Analytics for Enhancing Street Light Coverage of Public Spaces." American Association of Geographers Annual Meeting, San Francisco, CA, USA, March 29–April 2, 2016.
- 2015 X. Feng and S. Myint. "The Effect of Spatial Heterogeneity of Land Cover Types on Land Surface Temperatures Within Different Land Use Categories." American Association of Geographers Annual Meeting, Chicago, IL, USA, April 21–25, 2015.
- 2014 A. T. Murray and X. Feng. "Street Light Coverage Optimization." 61st North American Meetings of the Regional Science Association International, Washington, DC, USA, November 12–15, 2014.

- 2014 X. Feng. "Understanding the Effect of Neighboring Land Cover Pattern on Land Surface Temperature of a Central Building Patch in Beijing." American Association of Geographers Annual Meeting, Tampa, FL, USA, April 8–12, 2014.
- 2011 X. Feng, S. Du, and H. Shu. "Spatial Regression Analysis in Hemorrhagic Fever with Renal Syndrome (HFRS) in China." IEEE International Conference on Spatial Data Mining and Geographical Knowledge Services and 8th Beijing International Workshop on Geographical Information Science, Fuzhou, China, July 6–8, 2011.

Honors and Awards

- TRELIS Fellow, selected through a competitive process to participate in an intensive program of professional development for academic women in the geospatial sciences. Program funded by the US National Science Foundation (Award #1660400). 2022
- Finalist, William L. Garrison Award for Best Dissertation in Computational Geography, American Association of Geographers. 2021.
- Excellence in Research Award, Department of Geography, University of California, Santa Barbara. 2019.
- 2nd Place, Student Paper Competition, Geographic Information Science and Systems Specialty Group (GISS-SG), American Association of Geographers Annual Meeting, Washington, DC, April 3-7. 2019.
- 2nd Place, Tiebout Prize, Best Graduate Student Paper Award Western Regional Science Association 57th Annual Meeting, USA, February 11-14. 2018.
- Dangermond Travel Grant, Department of Geography, UCSB. Winter 2018, Spring 2018, Spring 2017, Fall 2017, and Fall 2016.
- Lounsbury Student Travel Fellowship, Arizona State University. 2014.
- Second Prize Scholarship, Peking University, China. 2013.
- Wusi Individual Scholarship, Peking University, China. 2012.
- First Prize Scholarship, Peking University, China. 2012.
- 2nd Place, Best Student Paper Award, IEEE International Conference on Spatial Data Mining and Geographical Knowledge Services, Fuzhou, China, July 6-8. 2011.
- Excellent B. S. Dissertation, Hubei Province, China. 2010.

Courses Taught

- GEOG 6240 Seminar on Geographical Artificial Intelligence
- GIS 4833/5833 Environmental Spatial Modeling
- GIS 5253 GIS Application (Online Master's Program)
- GIS 2023 Introduction to Spatial Thinking and Computer Mapping
- GIS 2023 Online - Introduction to Spatial Thinking and Computer Mapping
- GEOG 6980 Research Doctoral Dissertation
- GEOG 5980 Research for Master's Thesis

- GEOG 5960 Directed Readings
- GEOG 5200 Internship in Geography
- GEOG 4953 Capstone

Students Advised

Doctoral (as Advisor)

- Bikram Parajuli (2022 – present) completed the general exam in Fall 2024 and is expected to graduate in Fall 2026; Recipient of the 2026 OU John Michael Caldwell Award for Excellence in Graduate Student Teaching. First Place in 2025 OU Graduate & Postdoctoral Research and Scholarly Activity Day Poster Competition. Recipient of 2025 OU Ralph & Margaret Olson Geography Scholarship.
- Chenchun Wang (2024 – present) formed the dissertation committee in Spring 2026 and advanced to candidacy in May 2026; Recipient of the 2026 OU Standley Award for Outstanding Graduate Student Publication.
- Olawale Rowland (2025 – present)

Master's (as Advisor)

- Oyinkansola Kehinde Tella (2026 – present)
- M. Shahriyar Parvez – graduated in 2024; currently a Ph.D. student at the University of Connecticut. Recipient of 2025 AAG Marble Fund Award for Innovative Master's Research in Quantitative Geography.

Doctoral (as Committee Member)

- Joshua Wimhurst – graduated in 2023.
- Chinedu Nsude – graduated in 2026.

Doctoral (as External Representative)

- Kevin Thiel – graduated in 2025.
- Elizabeth Spicer (2025 – present)

Master's (as Committee Member)

- David Conn – graduated in 2022.

Undergraduate (as Capstone Advisor/Supervisor)

- Tyler Franklin
- Cole Howard
- Christian Rocha
- J. D. Love

Professional Service

Professional Society Service

Boards: Spatial Analysis and Modeling Specialty Group, American Association of Geographers, Board of Officers (2022–2025).

Funding/Grant Reviewer

NSF Graduate Research Fellowship Program (GRFP) 2024

NSF Human-Environment and Geographical Sciences Program (HEGS) 2025

Session Organizer and Chair

American Association of Geographers Annual Meeting, Spatial Analysis and Optimization Sessions, Session Organizer and Chair (2022, 2023, 2024, 2025, 2026)

North American Meetings of the Regional Science Association International, Location and Spatial Analysis Sessions, Session Chair (2024)

Journal Reviewer

Annals of GIS

Building and Environment

Cartography and Geographic Information Science

Cities

Computational Urban Science

Computers, Environment and Urban Systems

Construction and Building Materials

Environmental Management

Environment and Planning B: Urban Analytics and City Science

Geographical Analysis

IEEE Access

International Journal of Applied Earth Observation and Geoinformation

International Journal of Geo-Information

International Journal of Geographical Information Science

ISPRS International Journal of Geo-Information

Journal of Geographical Systems

Journal of Transport Geography

LEUKOS

Nature Communications

Papers in Regional Science

Socio-Economic Planning Sciences

Sustainable Cities and Society

Transportation Research Part E: Logistics and Transportation Review

Transactions in GIS

University and Department Committees

University of Oklahoma, Department of Geography and Environmental Sustainability

Award and Scholarship Committee (2025-Present)

Graduate Committee (2023-Present)

Faculty Search Committee

Director of the Center of Spatial Analysis (2021-2022)

Earth System Science Position (2024-2025)

Colloquium Committee (2021-2024)

Technology Committee (2021-2022)

University of Oklahoma, College of Atmospheric and Geographic Sciences
Cluster Hire Committee Focused on Community Resilience to Extreme Weather (2025-2026)

University of Oklahoma, Faculty Senate
Faculty Compensation & Benefits Committee, Faculty Senate (2025-2027)

University of California, Santa Barbara, Department of Geography (2016-2019)
Events Committee
Outreach Committee

Professional Society Membership

American Association of Geographers
Regional Science Association International
American Meteorological Society